



Origins of reduced winter rainfall retention on the Wye catchment above Belmont

A report prepared on behalf of Wye Salmon Association (WSA)

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Introduction

This report is concerned with the causes of the long-term increase in frequency of high flow events on the River Wye, as monitored by the EA gauging station at Belmont. It builds on two earlier reports on the topic.

The first report “Winter flows at Belmont” (Green, 2024b) looked at historical river discharge data and showed that the frequency of high flow events at Belmont increased dramatically during the period 1980-2000, being substantially constant before and after this period. In recent years, the river exceeds 4m three times as often as it did in the 1970s; for 5m exceedance, the factor is five times¹. Little of this appears to be attributable to climate change. Rather, it appears to be primarily driven by land management changes².

The second report “Rainfall retention on the Wye catchment above Belmont” (Green, 2024a) further examined rainfall patterns over the same period. It found that mean winter rainfall (Oct-Mar) has increased by 14% over the 50-year period 1973-2023. This increase was characterised simply by fewer dry days each year. There is no evidence supporting the idea that rainfall has become more intense on any given day. Over the same period, however, the mean river discharge has increased by 35%, or by 2.5x more than the increase in rainfall. Assuming that transpiration and evaporative losses are negligible in winter, the implication is that the catchment is retaining much less winter rainfall than it used to. Specifically, about half as much.

This then begs several questions: is winter retention a useful metric? If so, is falling retention specific to the Belmont catchment, or is it typical of a wider problem? And finally, what particular ‘land management’ changes might have driven it? Where was this water previously being retained? The objective of this report is to explore these questions.

Catchment geography

Topography and land use

It is useful at this point to examine some geographical aspects of the Belmont catchment. Figure 1 overleaf is a topographical map of the catchment (red outline). The dashed red outlines show some key sub catchments – the Ithon, Elan and Irfon – and the yellow line is the Welsh/English border. Also shown are NRW/EA gauging stations; red circles indicate volumetric flow measurement and triangles indicate height only stations. Each of the Ithon, Elan and Irfon have volumetric flow stations close to their Wye confluences. On the main stem, we find volumetric flow measurement on the upper catchment at Ddol Farm near Rhayader (upstream of the Elan confluence), then downstream at Erwood, before the final station at Belmont. Between these points, there are height-only stations at Builth Wells (below the Irfon confluence) and at Hay on Wye.

Most of the catchment by area lies upstream of Hay and much of this is high elevation. At Plymlimon, where the Wye begins, we find peaks around 700m elevation. Much of the Elan sub catchment lies above 500m and the ridge between the Elan and Irfon sub catchments peaks at over 600m. The elevation on the main stem at Rhayader is 200m, falling to 130m at Builth Wells. So, these western tributaries lose several hundred metres of elevation before they drain into the Wye. The Ithon sub catchment to the north-east also has 500m hills around its upland boundaries, but for the most part has a gentler topography.

¹ 4m is the level where low-lying land becomes flooded; above 5m takes us into “property damage” territory, with inundated fields all the way down to Monmouth and beyond.

² A view apparently shared by the Environment Agency.

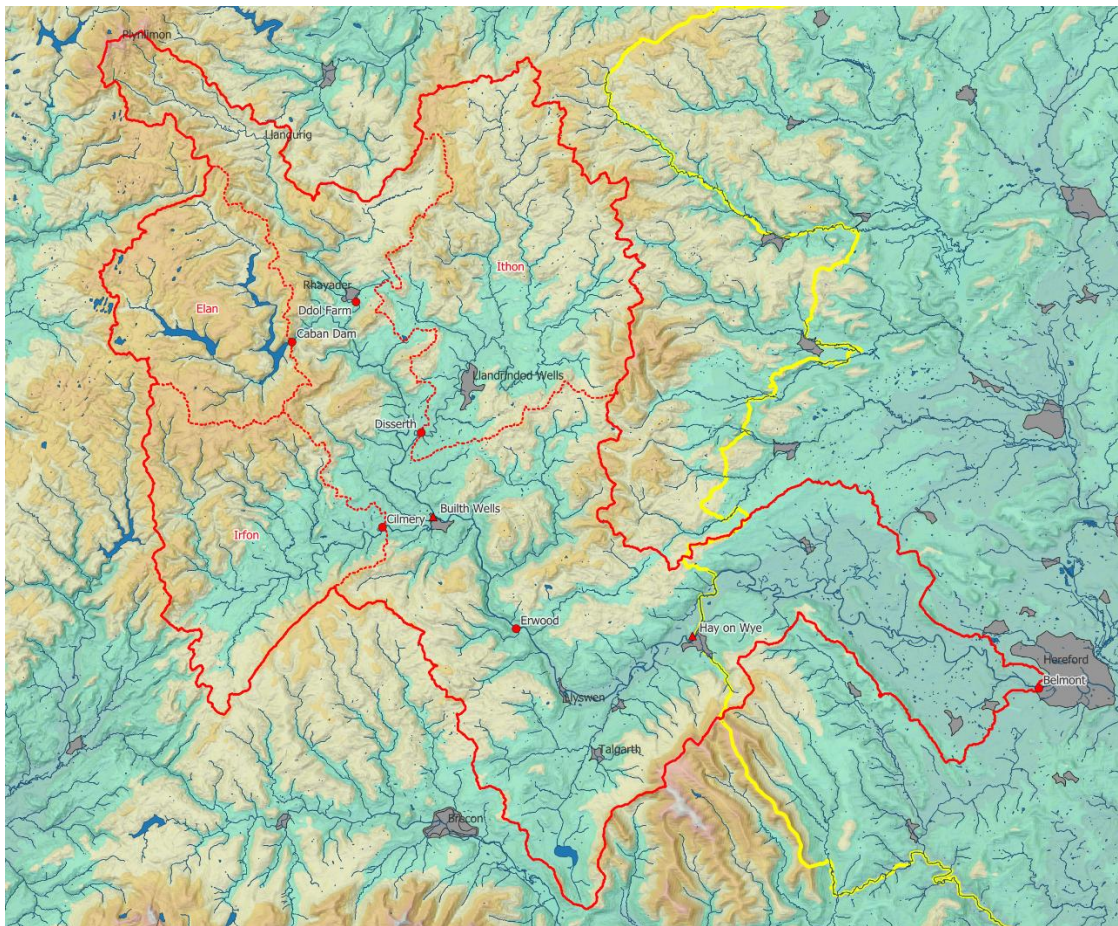


Figure 1: topographical map of the Belmont catchment

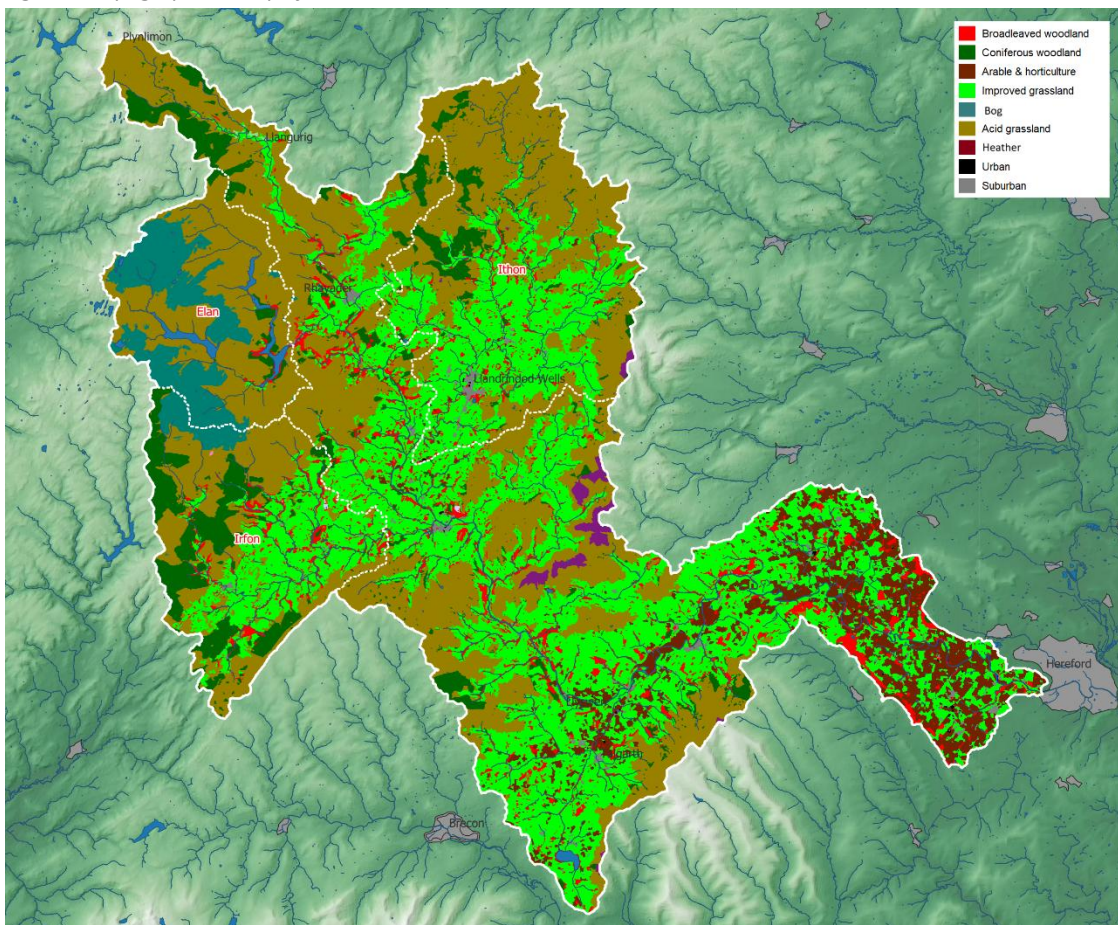


Figure 2: land use map of the Belmont catchment (CEH 2020)

From Builth, the main stem loses elevation rapidly down to Hay at approximately 70m elevation. Below Hay, the river becomes relatively more sedate, falling to 50m elevation at Belmont some 46km downstream. Note that the lakes within the Elan catchment provide drinking water to the Midlands, with the flow regime below Caban Dam consequently being 'unnatural'.

Figure 2 shows land use across the catchment. Below Hay (in Herefordshire) we find a significant amount of arable farming (brown), whereas the Powys lowlands are almost entirely improved pasture (light green). The boundary between improved pasture and acid grassland (light brown) roughly follows the contours, with land much above 300m generally being unimproved. Coniferous forestry (dark green) for the most part lies above 350m, with significant areas occurring in the upper Wye, Irfon and Ithon sub catchments. Note also the significant area of bog (teal) occurring in the Elan and Irfon sub catchments.

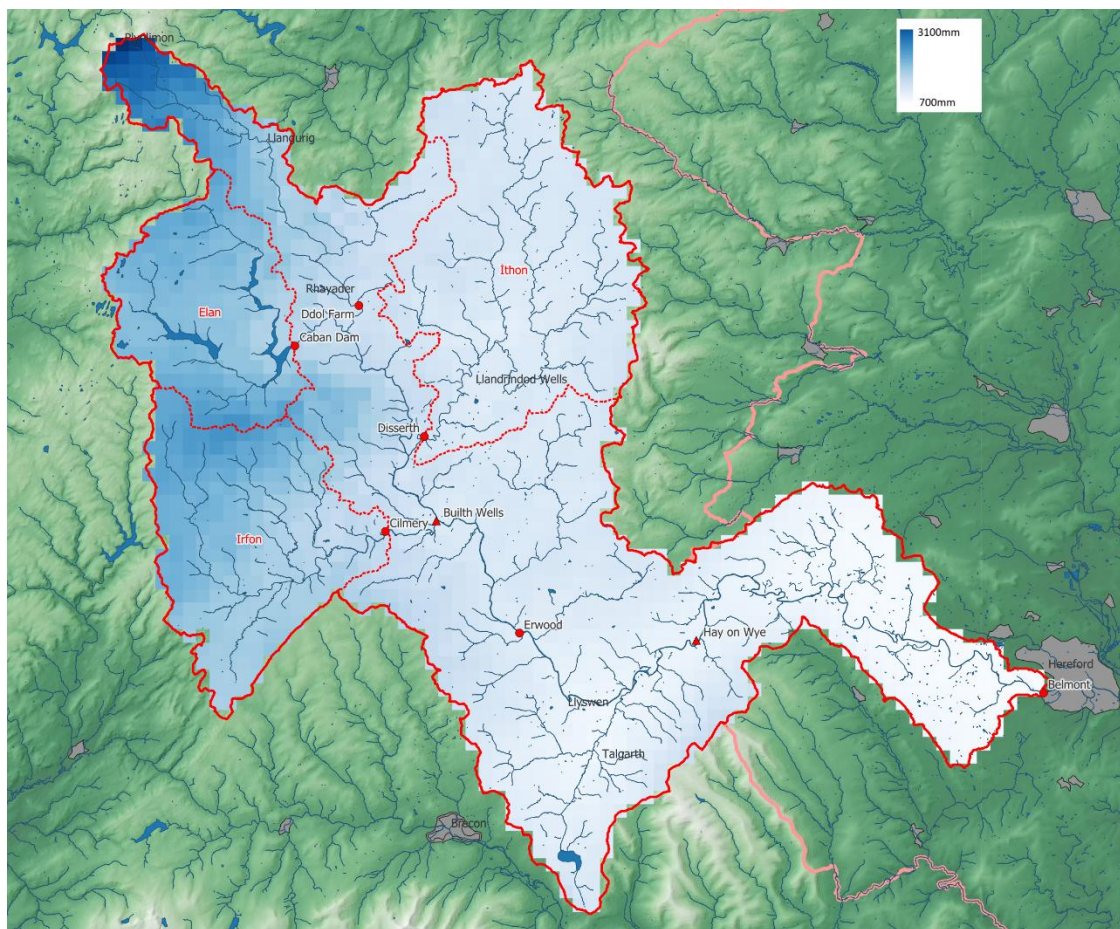


Figure 3: average annual rainfall (2015-19) on the Belmont catchment

Figure 3 above shows the average annual rainfall distribution on the Belmont catchment. There are massive contrasts here. The area immediately above Belmont receives around 700mm of rainfall annually. Upstream at Builth Wells, this increases to around 1000mm. However, the western hills are receiving significantly more than this. The ridge between the Irfon and Elan sub catchments catches around 2200mm and the Plynlimon area receives around 3000mm. Also of note is that the high ground to the north and east of the Ithon sub catchment doesn't exceed about 1200mm annual rainfall. This suggests that the western side of the catchment has a disproportionate impact on the amount of water flowing past Belmont.

We can clearly illustrate this point by examining flows at the different monitoring stations. Figure 4 is a much-simplified catchment schematic, containing the annual average volumetric flows across the Belmont catchment. Note that four fifths of the mean annual flow at Belmont is already in the river at

Erwood, 68km upstream. Furthermore, almost three quarters of the Belmont flow originates upstream of Builth Wells (Builth to Belmont circa 82km). Thus, the narrow strip of Herefordshire catchment lying downstream of Hay has a negligible effect on flows past Belmont. Belmont flows are primarily determined by how water flows off the Welsh uplands.

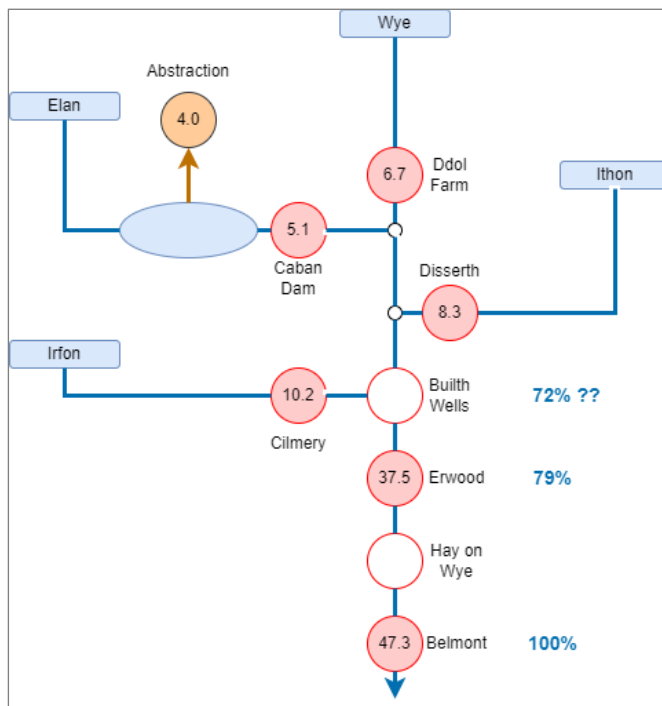


Figure 4: current annual average volumetric flowrates (m³/s) at each of the monitoring station (<https://nrfa.ceh.ac.uk/>)

A typical 5m flood

To reinforce this point, it is interesting to track a specific flood event propagating down the sub catchment. The most recent event at the time of writing resulting from a band of heavy rain passing through mid-Wales and central England on 05-Jan-25 following a few days of dry weather. By midnight, the gauge at Ddol Farm had increased from 0.7m to 1.8m, a rise of 1.1m (figure 5). Simultaneously, the Irfon at Cilmercy rose 2.5m.

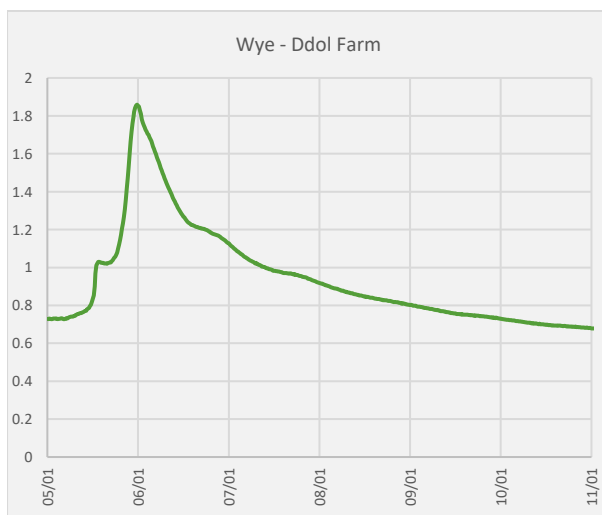


Figure 5: pulse at Ddol Farm

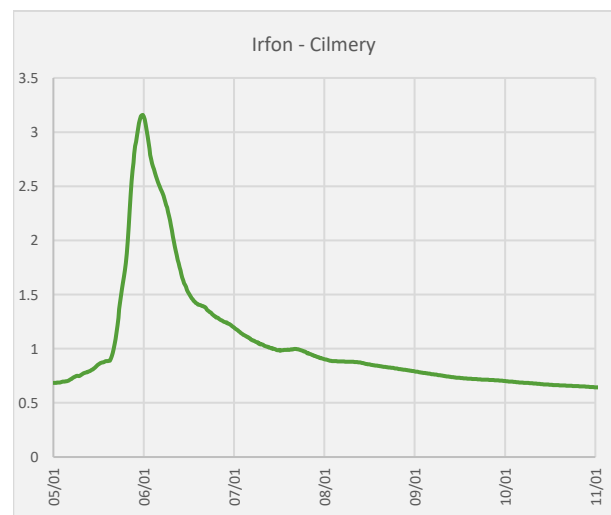


Figure 6: pulse at Cilmercy

Note the leading edge of these pulses. The Irfon for example went from 1m to 3m in approximately 10 hours. The fall was rapid too, essentially the whole event was over in 24 hours, with the river back to where it started in about 48 hours. The height at Ddol Farm behaved similarly. This was a very rapid

dumping of water, with limited retention. Figure 6 summarises how the pulse then propagated down river (time-delay in red). The peak at Builth Wells (figure 7) occurred 3 hours later, essentially being made up of Wye and Irfon water. The peaks from the Elan and Ithon occurred somewhat later, so they contributed to the tail of the pulse at Builth, but not to the height of the initial peak. Erwood peaked at 5.4m, corresponding to a volumetric flow in the region of 500 m³/s.

The rest is inevitable. Below Erwood we see a progressive increase in the pulse amplitude. This is not primarily extra water but is a simple consequence of reduced gradient. As the flow speed reduces on the lower gradients in Herefordshire, the height must increase to carry the same volume of water – at least until the banks overtop. Twenty-five hours after the pulse at Ddol Farm, the Wye at Belmont rose by over 4m, to peak at 5.4m (figure 8). The inflection (circled) is probably due to overtopping which limits the in-channel flow. The peak at Belmont thus corresponds to a volumetric flowrate over 400 m³/s – similar to, but actually slightly less than, that which flowed through Erwood, confirming the idea that this peak flow was driven by what occurred upstream of Erwood.

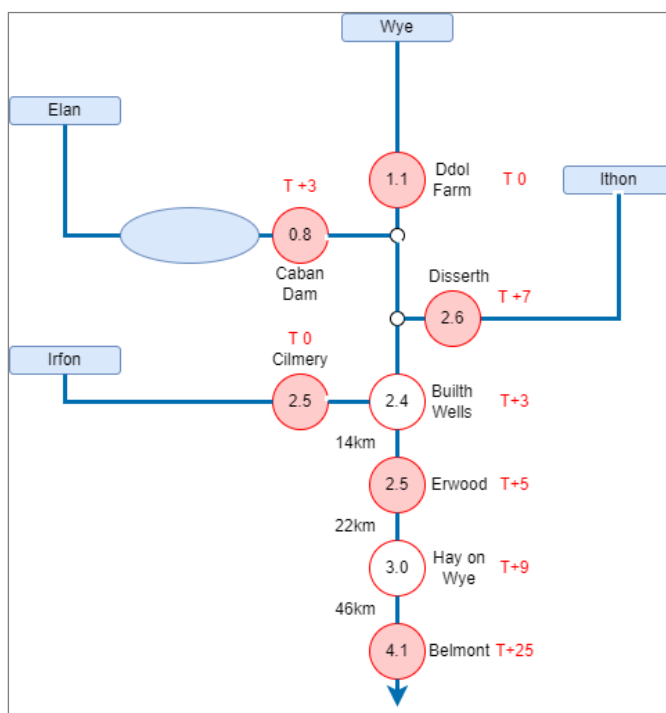


Figure 6: river height increases resulting from heavy rainfall on 05-01-2025. Numbers in red circles indicate height increase at each location. Legends in red indicate timing of the peak height relative to the peak at Ddol Farm.



Figure 7: pulse at Builth Wells

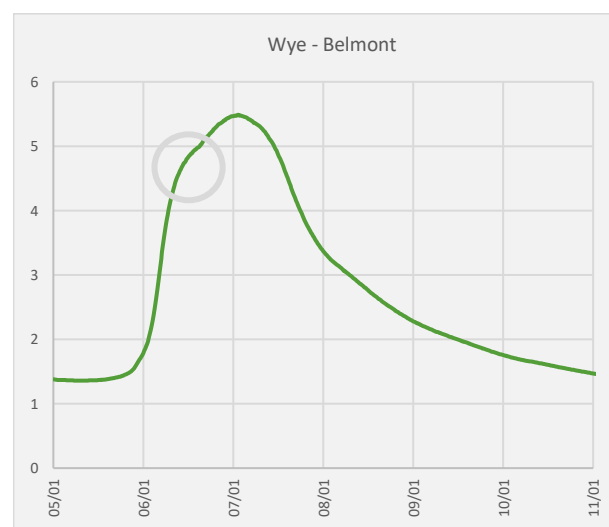


Figure 8: pulse at Belmont

Winter retention

Magnitude

In my earlier report (Green, 2024a), we found that the amount of winter rainfall retained in the catchment above Belmont (i.e. winter rainfall minus river discharge) had fallen significantly between the 1970s and today. In the 1970s the catchment retained on average $395 \times 10^6 \text{ m}^3$, whereas today it retains only $223 \times 10^6 \text{ m}^3$ – a fall of $172 \times 10^6 \text{ m}^3$, or 44%. What does this figure represent?

$172 \times 10^6 \text{ m}^3$ of water is enough to take the Wye at Belmont from its median flow (Q_{50}) to 5m height for about 7 days each winter, so it represents a very significant volume of water. Perhaps of more relevance is that it would increase a 'normal' 3.5m pulse to 5m for about 15 days per winter – which is the right order of magnitude for what we are seeing in the change in flood frequency and duration between the two periods. Thus, it is entirely plausible that reduced retention is indeed an indicator of more rapid run-off during heavy rainfall. Moreover, if the non-retained water is now finding its way rapidly to the river, then it would clearly be associated with the rapid-rise, high peak events which have become the new normal. This certainly seems to be borne out by the events of early January described above.

Comparing other catchments

This all begs the question of whether reduced winter retention is unique to the Wye above Belmont. I have calculated the same statistic for some relevant Wye sub catchments and for some neighbouring rivers. The results are listed in table 1.

Catchment	Change in winter retention
Wye above Belmont	-44%
Irfon above Cilmerly	-78%
Ithon above Disserth	-4%
Wye above Ddol Farm	-36%
Llynfi above Three Cocks	-18%
Lugg above Lugwardine	36%
Lugg above Butts Bridge	31%
Teme above Tenbury Wells	-13%
Usk above Chainbridge	-12%
Ystwyth above Pont Llotwyn	-75%

Table 1: change in retention of winter rainfall between 1973 and 2023, calculated for a selection of sub catchments in and around the Wye Belmont catchment.

The Wye Belmont catchment is in no way unique in having lost retention capacity over the last fifty years; but we find a huge range within these results. Of particular note are the Irfon, Ystwyth and Wye above Ddol Farm, all with very large retention falls, and geographically in close proximity. The Usk and Teme also show falls but of smaller magnitude. The Ithon has fallen a little, and the adjacent upper Lugg has actually increased.

In order to find out what could be driving these retention numbers, it is helpful to correlate the changes against topography and land use for each sub catchment. In figure 9a, we can see a strong negative-slope correlation ($R^2=0.48$) with the area of land above 450m in each catchment. The reduction in winter rainfall retention appears to be strongly driven by the upland areas. We now look at land use³. But bear in mind that some land uses themselves are restricted to high elevation land and will necessarily co-correlate. Nevertheless, the coniferous woodland correlation (figure 9b) is interesting in that it shows a significantly stronger correlation ($R^2=0.59$) than elevation alone. This rather suggests that this is a significant result. Looking at other upland land uses, we find a significant positive-slope correlation with heather ($R^2=0.49$) and no discernible correlation with grassland types. The latter implies

³ Calculated using the 2000 CEH 25m raster dataset.

that changes in retention are not being driven by sheep farming practices. Arable land has a positive-slope correlation ($R^2=0.34$), possibly suggesting that it's better than compacted grassland at absorbing rainwater. Thus, the increase in retention found in the Lugg sub catchment may possibly reflect an increase in arable farming since the 1970s.

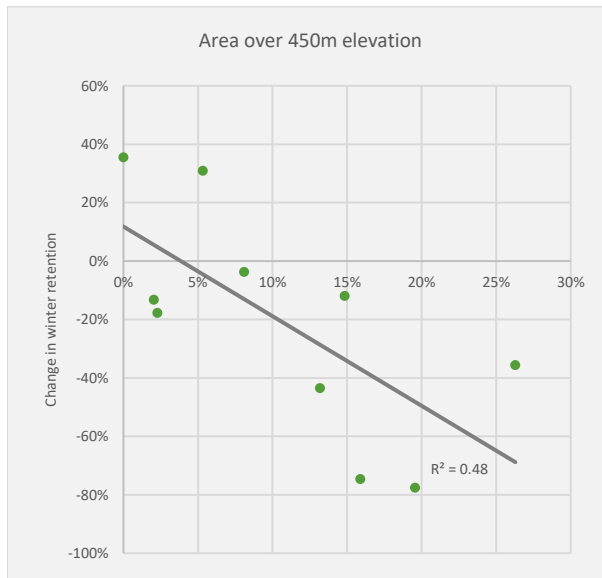


Figure 9a: correlation between fall in winter retention and fraction of land over 450m elevation



Figure 9b: correlation between fall in winter retention and fractional area of coniferous woodland

To summarise, we find a large variation in changes to winter rainfall retention (1973 to 2023) within the 11 catchments, ranging from -78% on the Irfon to +36% on the Lugg. Reductions in retention correlate strongly with area of land over 450m and with coniferous woodland cover.

Historical land use data

The satellite data made available by CEH does not pre-date 1990, so we must go to other sources to understand what changed in the upper reaches of the Wye catchment between 1980 and 2000 (this being the period when flooding patterns shifted dramatically). Figure 8 summarises some information published by Welsh Government but originating from DEFRA which approximately covers our period of interest.

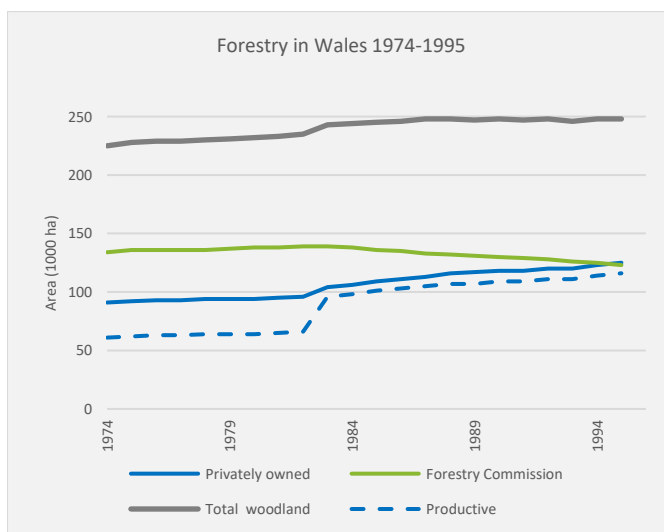


Figure 10: changes in the Welsh forestry sector 1974 to 1995.

Changes to legislation (Forestry Act, 1981) drove a progressive shift in forest ownership away from the Forestry Commission (now replaced by NRW in Wales) and into private ownership. More importantly, it enabled a significant and immediate switch to productive forestry (logging). Between 1974 and 1995, figure 10 shows that there was a 37% increase in privately owned forests across Wales and a 90% increase in productive forestry.

Unfortunately, I can find no specific data for Powys, or the pre-existing counties, far less for the Wye catchment. However, Ordnance Survey mapping can tell us a lot. Figure 11 shows a typical area of the mid-Wales uplands, in this case the upper reaches of the Irfon sub catchment (to the east of the red line) and of the Ystwyth (to the west of the red line). The Irfon can be seen snaking its way north to south, via Carreg yr Adar and Camddwr Bleiddiad. The area is heavily covered by coniferous woodland, over the summits of the round-topped hills and often down to the valley bottoms. Moreover, this area is not cherry-picked; it is typical of the headwaters of the Irfon, Wye and Ystwyth, and quite possibly of many more Welsh rivers.

In figure 12 we can see the same area of OS mapping from 1964. The difference is stark, not a conifer in sight. The entire area of the map used to be covered in open heathland, crucially including large areas of bog on the tops of the hills. Comparison with figure 11 shows us that huge areas of this previously absorbent land have since been converted to forestry – undoubtedly incorporating rapid drainage to surface water. Moreover, note the steep gradients on the valley sides. Drainage of this land is inevitably rapid if rain is not retained at its point of impact. Even in the valley bottoms, the streams are steep up here. The Irfon itself loses about 100m of elevation as it flows the 3km across this map area. This is energetic flow.

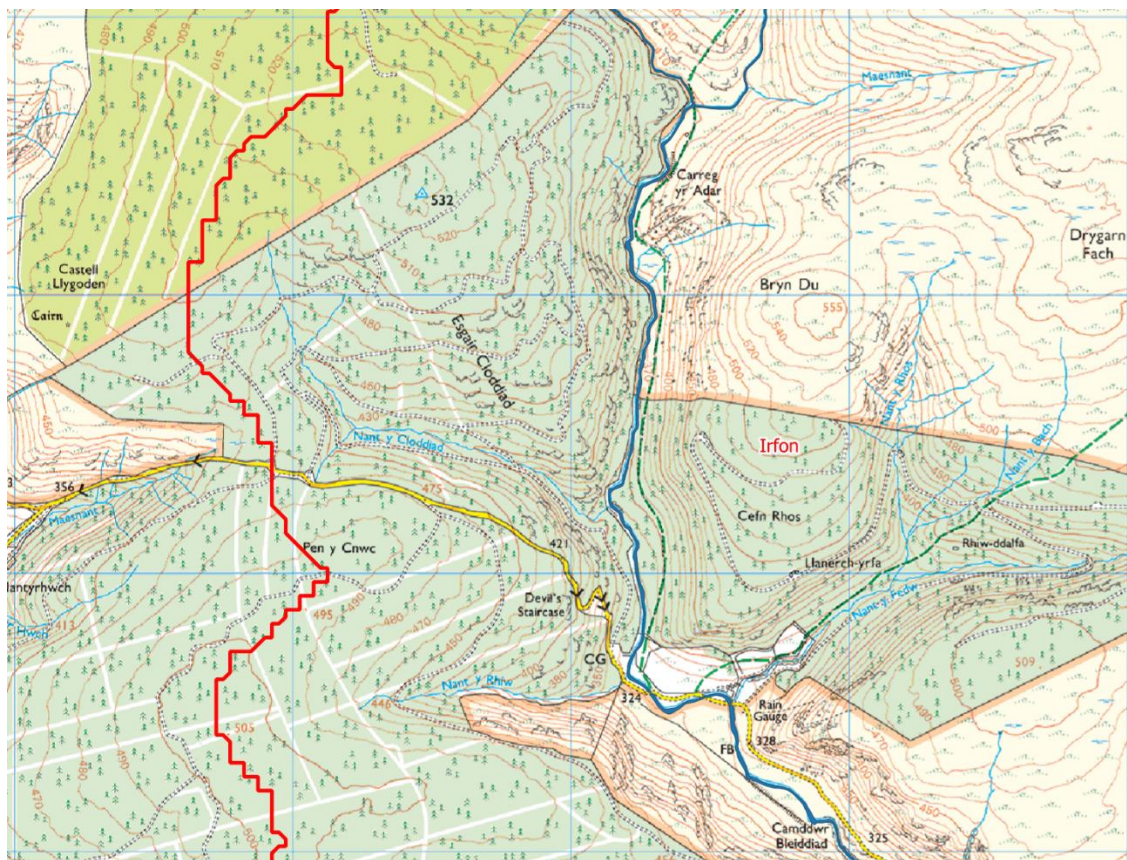


Figure 11: 12 km² of the Irfon/Ystwyth catchment in 2023



Figure 12: the same 12km² as shown in figure 11, circa 1964 (OS mapping)

This now appears to be making sense. It is known that logging requires access for heavy machinery and hence is not compatible with boggy ground. The clear inference is that logging activities have necessitated drainage of the land on the hill tops. Thus, land which used to behave like a giant sponge, retaining water for weeks and months, has been engineered to discharge as rapidly as possible down to the river. Moreover, the areas surrounding the headwaters of the Irfon, Elan, Wye and Ystwyth (the high retention loss rivers) all contain large areas of coniferous forest.

Conclusions

Recapping the two earlier reports, we found that:

- Bank-overflowing 5m exceedances at Belmont occurred two or three times per decade in the 1960s and 70s; since 2000, they have occurred several times each year. This 5x change in frequency was not progressive but occurred relatively suddenly between 1980 and 2000.
- Winter (Oct-Mar) rainfall has increased monotonically by 14% between 1973 and 2023, presumably due to climate change. There are now more wet days than there used to be. We found no evidence to support the idea that the sub catchment receives more intense rainfall (despite the modern practice of naming storms).
- Over the same period, the mean river discharge has increased by 35%, 2.5x the increase in rainfall. This implies a reduction of 44% in winter rainfall retention within the sub catchment.

In this report, we have seen that:

- On an annual mean basis, four fifths of the water flowing past Belmont originates upstream of Erwood. The amount originating within Herefordshire is negligible. Flows through Builth and Erwood are biased toward the western side of the catchment where the rainfall is significantly higher.

- Flow pulses from the western side of the catchment are characterised by very rapid rise and fall. The resulting high peak flows thus accumulating at Builth Wells are directly driving the 5m floods downstream at Belmont. The peak volumetric flow passing Erwood on 06-Jan-25 was essentially the same as the peak flow through Belmont 20 hours later. Thus, any mitigation for winter flooding in the middle Wye must firstly address these 'rapid discharge' characteristics in the headwaters.
- The reduction of winter rainfall retention within the Wye Belmont catchment represents a significant volume of water. It is equivalent to 15 days of 3.5m flow at Belmont being raised to 5m.
- Many other catchments are found to have reduced winter rainfall retention too, notably the Ystwyth. Within the Belmont catchment, the Irfon and Wye above Ddol Farm are severely affected. The amount of reduction between 1973 and 2023 is shown to correlate with land over 450m elevation and strongly with forestry. It does not correlate with sheep farming.
- Commercial forestry expanded rapidly in Wales following changes introduced by the 1981 Forestry Act. By 1995, commercial forestry had increase by 90% across Wales. OS mapping shows, as an exemplar, that large areas of the upper Irfon catchment were transformed from mostly open heath and bog in 1964 to hydrologically significant areas of conifer plantation now. It is known that commercial forestry operations require drainage of these previously boggy upland areas. The high gradients present in the valley sides then dictate that this drainage is rapid, driving the rapid-rise, high peak flow surges which then propagate through the length of the river Wye.

Whilst the above analysis does not prove a direct causal link, it does establish a very compelling correlation suggesting that the frequency and severity of winter flooding events at Belmont are primarily a consequence of the replacement of natural boggy moorland by commercial forestry in the Welsh headwater areas⁴. It is not being significantly driven by climate change to date. The draining of upland bogs and heath must surely be reversed as part of any habitat restoration project or climate change mitigation program. The bogs need to be reinstated.

References

Green, G. (2024a). *Rainfall retention on the Wye catchment above Belmont*.

Green, G. (2024b). *Winter flows at Belmont*.

Acknowledgements

Catchment maps make use of OS data © crown copyright 2022, [OS Open Zoomstack](#) | [OS Products](#) (ordnancesurvey.co.uk) and are prepared using [QGIS](#) open-source software.

Rainfall and river discharge data is primarily from the [CEH National River Flow Archive](#). 2025 discharge data comes from the [DEFRA Hydrology Data Explorer](#) and the [NRW River Levels Explorer](#).

⁴ Such a conclusion is probably not contentious – just not widely understood. DEFRA/EA have researched the [impacts of land use on flooding](#) over many years. Impacts from forestry and from bog drainage are well known. A report from their *Making Space for Water* project in the [Peak District](#) makes the point explicitly. Closer to home, Wye & Usk Foundation refer to the [impact of forestry drains](#) on flood patterns. Also, within the current NRW [Upper Wye Catchment Restoration Project](#), we find a 'Slow the flow' workstream. Sadly though, NRW are somewhat reticent about the root causes of the problem the workstream hopes to mitigate.